

Cambridge International AS & A Level

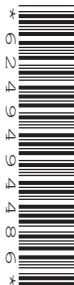
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FURTHER MATHEMATICS

9231/31

Paper 3 Further Mechanics

May/June 2022

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

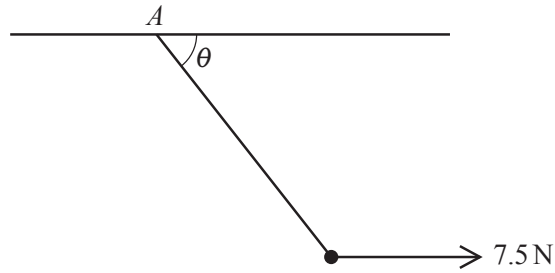
- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2} .

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

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A particle of weight 10 N is attached to one end of a light elastic string. The other end of the string is attached to a fixed point A on a horizontal ceiling. A horizontal force of 7.5 N acts on the particle. In the equilibrium position, the string makes an angle θ with the ceiling (see diagram). The string has natural length 0.8 m and modulus of elasticity 50 N .

- (a) Find the tension in the string. [2]

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- (b)** Find the vertical distance between the particle and the ceiling. [3]

[illegible]

[5]

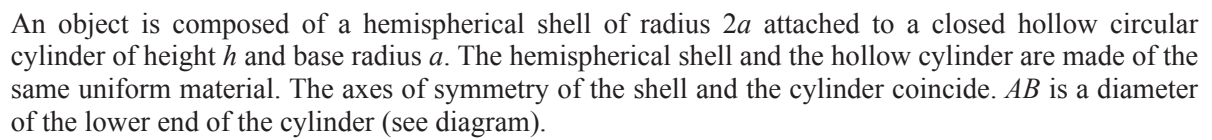
[illegible]

- 3** A particle P is moving in a horizontal straight line. Initially P is at the point O on the line and is moving with velocity 25 m s^{-1} . At time $t \text{ s}$ after passing through O , the acceleration of P is $\frac{4000}{(5t+4)^3} \text{ m s}^{-2}$ in the direction PO . The displacement of P from O at time t is $x \text{ m}$.

Find an expression for x in terms of t .

[5]

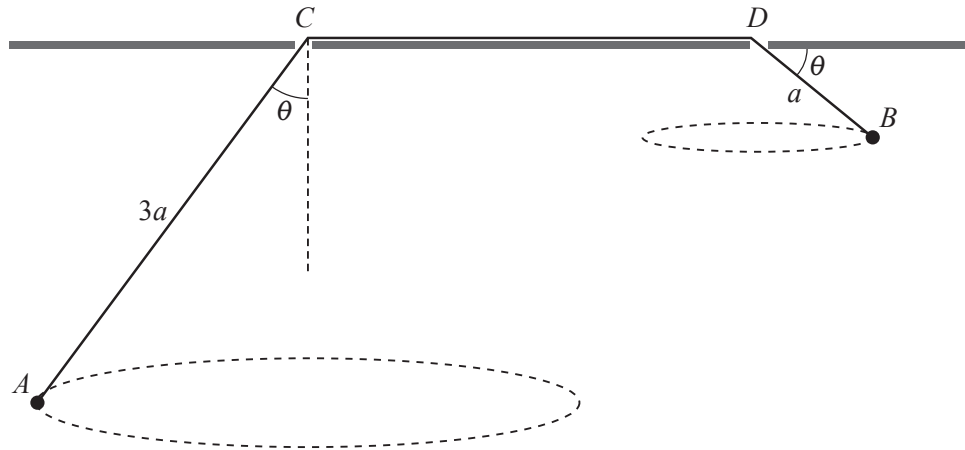
[illegible]



- [illegible]

The object is placed on a rough plane which is inclined to the horizontal at an angle θ , where $\tan \theta = \frac{2}{3}$. The object is in equilibrium with AB in contact with the plane and lying along a line of greatest slope of the plane.

- (b) Find the set of possible values of h , in terms of a . [4]



A light inextensible string AB passes through two small holes C and D in a smooth horizontal table where $AC = 3a$ and $DB = a$. A particle of mass m is attached at the end A and moves in a horizontal circle with angular velocity ω . A particle of mass $\frac{3}{4}m$ is attached to the end B and moves in a horizontal circle with angular velocity $k\omega$. AC makes an angle θ with the downward vertical and DB makes an angle θ with the horizontal (see diagram).

Find the value of k .

[7]

[illegible]

[illegible]

After the collision, the kinetic energy of A is equal to the kinetic energy of B .

- (b) Given that $\tan \alpha = \frac{2}{3}$, find the possible values of k . [5]

[illegible]

[illegible]

(b) Find the value of T .

[4]

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(c) Find the horizontal and vertical displacements of the particles from O when they collide.

[3]

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